

# DATA MANAGEMENT 2 — GROUP COURSEWORK

We will work independently and connect everything at the end.

## IMPORTANT RULE

Each member works ONLY in their assigned folder.

```
coursework/  
├── oracle/  
├── plsql/  
├── mongodb/  
├── application/  
└── documentation/
```

- Member 1 → `oracle/`
- Member 2 → `plsql/`
- Member 3 → `mongodb/`
- Member 4 → `application/`
- `documentation/` → everyone contributes later

Do NOT delete or modify another member's folder.

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## MEMBER 1 — ORACLE DATABASE

### Folder

`oracle/`

### Your job

Build the complete relational database using Oracle.

### Tasks

- Read the coursework requirements.
- Identify required entities/tables.
- Design the relational database.
- Create tables.

- Define primary keys.
- Define foreign keys.
- Add required constraints.
- Insert sample/test data.
- Write required SQL queries.
- Test all SQL.
- Keep the SQL scripts organized.

## Files

Create:

```
oracle/
├── create_tables.sql
├── insert_data.sql
└── queries.sql
```

If necessary, add:

```
oracle/
├── views.sql
└── README.md
```

## Software

Use:

- Oracle Database
- Oracle SQL Developer

## When finished

Make sure another person can recreate the database by running your SQL scripts in order.

## Git

Before starting:

```
git pull
```

After completing a small task:

```
git add oracle/
git commit -m "Add Oracle database tables"
git push
```

Other good commit messages:

Add Oracle constraints  
Add Oracle sample data  
Add required SQL queries  
Fix Oracle foreign keys

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## MEMBER 2 — PL/SQL

### Folder

plsql/

### Your job

Implement the required PL/SQL functionality.

### Tasks

Create and test:

- Procedures
- Functions
- Triggers
- Cursors, if required
- Exception handling
- %TYPE
- %ROWTYPE
- Variables
- IF/ELSE
- Loops
- Other PL/SQL requirements in the coursework

### Files

plsql/

```
|— procedures.sql  
|— functions.sql  
|— triggers.sql  
|— cursors.sql  
|— exceptions.sql
```

Only create files that are actually required.

### Software

Use:

- Oracle Database
- Oracle SQL Developer

## Important

Your PL/SQL will eventually use the Oracle database created by Member 1.

Since we are working independently, use the **table/entity names specified by the coursework**. Do not randomly invent a completely different database structure.

## Git

Before starting:

```
git pull
```

After completing work:

```
git add plsql/  
git commit -m "Add PL/SQL procedures"  
git push
```

Other examples:

Add PL/SQL functions  
Add database triggers  
Add exception handling  
Add cursor implementation  
Fix PL/SQL procedure

---

# MEMBER 3 — MONGODB

## Folder

mongodb/

## Your job

Build the MongoDB part of the coursework.

## Tasks

- Identify required MongoDB data.

- Design collections.
- Create database/collections.
- Insert sample documents.
- Write required queries.
- Implement updates.
- Implement deletes.
- Implement aggregation if required.
- Test everything.
- Keep scripts organized.

## Files

```
mongodb/  
├── create_collections.js  
├── insert_data.js  
├── queries.js  
├── updates.js  
└── aggregation.js
```

Only keep files that are required.

## Commands to use

```
insertOne()  
insertMany()  
find()  
findOne()  
updateOne()  
updateMany()  
deleteOne()  
deleteMany()  
aggregate()
```

## Software

Use:

- MongoDB
- MongoDB Compass
- mongosh if needed

## Git

Before starting:

```
git pull
```

After completing work:

```
git add mongodb/  
git commit -m "Add MongoDB collections"  
git push
```

Other examples:

```
Add MongoDB sample documents  
Add MongoDB queries  
Add MongoDB update operations  
Add MongoDB aggregation
```

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## MEMBER 4 — APPLICATION

### Folder

application/

### Your job

Build the application/interface.

The database connections can be added later during integration.

### First build the application structure

Create:

- Main page
- Navigation
- Dashboard
- Forms
- Tables
- Search
- Add/Edit/Delete interfaces
- Validation
- Error messages
- Other pages required by the coursework

### Files/structure

The exact structure depends on the technology, but something like:

```
application/  
├── src/  
├── pages/
```

```
|— components/  
|— assets/  
|— README.md
```

## Software

Use the technology required/appropriate for the coursework, for example:

- VS Code
- Python / Java / C# / JavaScript
- Git

## IMPORTANT

You do NOT need to wait for the databases to be finished.

Build the UI first using sample/mock data.

Later we will connect:

Application

↓

Backend/API

```
|— Oracle  
|— MongoDB
```

## Git

Before starting:

git pull

After completing work:

git add application/

git commit -m "Create application structure"

git push

Other examples:

Add dashboard

Add employee form

Add search functionality

Add application validation

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# DOCUMENTATION — EVERYONE

Folder:

documentation/

Nobody needs to complete this immediately.

During the project, save:

- Screenshots
- ERD
- Database diagrams
- Query results
- PL/SQL results
- MongoDB results
- Application screenshots
- Testing evidence

Eventually the documentation should cover:

1. Introduction
2. Requirements
3. System design
4. ERD
5. Oracle database
6. SQL
7. PL/SQL
8. MongoDB
9. Application
10. Integration
11. Testing
12. Results
13. Conclusion
14. References

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# GIT RULES — EVERYONE

## 1. Before working

Always do:

git pull



## 2. Work only in your folder

Member 1 → oracle/

Member 2 → plsql/

Member 3 → mongodb/

Member 4 → application/

## 3. Don't make huge commits

Bad:

Final coursework

Good:

Add Oracle employee table

Add MongoDB sample data

Add PL/SQL trigger

Add application dashboard

## 4. After finishing a task

git add .

git commit -m "Describe what you actually changed"

git push

## 5. Don't commit passwords

Never put database passwords/API keys in GitHub.

Use `.env` for private configuration and add it to `.gitignore`.

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# SOFTWARE CHECKLIST

### Member 1

- Oracle Database
- Oracle SQL Developer
- Git
- GitHub

## Member 2

- Oracle Database
- Oracle SQL Developer
- Git
- GitHub

## Member 3

- MongoDB
- MongoDB Compass
- mongosh
- Git
- GitHub

## Member 4

- VS Code
  - Required programming language/framework
  - Git
  - GitHub
  - Database drivers/connectors later
- 

# WORKING ORDER

Everyone can start now.

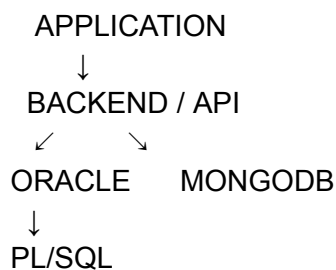
Member 1 → Oracle

Member 2 → PL/SQL

Member 3 → MongoDB

Member 4 → Application UI

We will connect everything later:



After all four parts are completed:

1. Connect application to Oracle.

2. Connect application to MongoDB.
3. Connect required PL/SQL functionality.
4. Test CRUD operations.
5. Fix compatibility issues.
6. Complete documentation.
7. Test the complete system.
8. Prepare presentation/demo.

## MOST IMPORTANT

Do not wait for other members to start.

Start your assigned work now.

But follow the **coursework requirements exactly** so that the four parts can be integrated later.